2/9/26.

EXPERIMENT - 7

Sliding Window Protocol.

Aim: To write a program to implement flow control at data link layer using sliding window protocol simulate the flow of frames from one node to another

Sender code:

```
import time
import os
print ("=" * 55)
print ("Sliding window protocol - sender")
window_size = int (input ("Enter window size"))
if (window_size <=0 ;
        print ("Invalid window size;")
        exit()
message = input ("Enter text message :")
if not message :
        print ("Message cannot be empty!")
        exit()
frames = []
for i, character in enumerate (message);
        frames.append ({"frames_no": i,
                        "data" : character
                        })
print ("Frames created:")
```

```
total_frames = len(frames)
base = 0
while base < total_frames:
    end = min(base + window_size, total_frames)
    print("sending frames {base} to {end}")
    for i in range(base, end):

try:
    with open("receiver_buffer.txt", "r") as file:
        response = file.readline().strip()
except FileNotFoundError:
    print("Receiver buffer not found")
parts = response.split()
print("All frames transmitted successfully")
print("Original message {message}")
```

Output:

```
SLIDING WINDOW PROTOCOL SENDER
Enter window size: 3
Enter the message: HELLO
```

Sender buffer:

```
Frame no: 0 DATA: H
Frame no: 1 DATA: E
Frame no: 2 DATA: L
Frame no: 3 DATA: L
Frame no: 4 DATA: O
```

Receiver node:

```
import os
SENDER_FILE = "sender_buffer.txt"
RECEIVER_FILE = "receiver_buffer.txt"
expected_frame = 0

print(f"{}. ii; All frames received correctly}")
print(f"Ack: {expected_frame}")
```

OUTPUT:

Frames received

frame no: 0 Data: H

frame no: 1 Data: E

frame no: 2 Data: L

All frames received correctly

ACK: 3

RESULT:

Thus, the sliding window protocol was implemented successfully using python and flow of frames b/w sender and receiver was simulated using ACK / NACK.